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Comparison of modeling methods for Loss Given Default

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ABSTRACT

We compare six modeling methods for Loss Given Default (LGD). We find that non-parametric methods (regression tree and neural network) perform better than parametric methods both in and out of sample when over-fitting is properly controlled. Among the parametric methods, fractional response regression has a slight edge over OLS regression. Performance of the transformation methods (inverse Gaussian and beta transformation) is very sensitive to ε, a small adjustment made to LGDs of 0 or 1 prior to transformation. Model fit is poor when ε is too small or too large, although the fitted LGDs have strong bi-modal distribution with very small ε. Therefore, models that produce strong bi-model pattern do not necessarily have good model fit and accurate LGD predictions. Even with an optimal ε, the performance of the transformation methods can only match that of the OLS.

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1. Introduction

The expected credit loss rate on any pool of debt over any given horizon can be expressed as the product of probability of default (PD) and Loss Given Default (LGD). Therefore, LGD is one of the two determining factors of the premium of risky bonds, credit default swap spreads, and credit losses. It is also one of the key parameters in the Basel II framework that are used to calculate banks' regulatory capital requirements.[1] In spite of its obvious importance, the number of studies on LGD has been relatively low, although growing in recent years. Most of the LGD studies focus on investigating the importance of various factors that affect LGD, for example, contract characteristics, borrower characteristics, industry conditions, and macroeconomic conditions.[2] Very few studies of LGD explore the alternative modeling methodologies.

LGD modeling methods are of interest because the distribution of LGD can be very different from the normal distribution, which is behind most of the commonly used statistical models. First, LGDs for corporate exposures tend to have a bi-modal distribution (for example, Schuermann (2004)). That is, the LGD is either very high or very low. Second, LGD is bounded between 0 and 1, while theoretically the predicted values from the ordinary least squares (OLS) regression can range from negative infinity to positive infinity. Despite this unconventional distribution, most studies on LGD continue to use the OLS regression.[3] Some studies take into consideration the LGD boundaries when choosing an LGD modeling method. For example, Dermine and Neto de Carvalho (2006), Bellotti and Crook (2007), and Bastos (2010) use fractional response regression, Hu and Perraudin (2002) use inverse Gaussian regression, and the Moody's LossCalc 2.0 (Gupton and Stein (2005)) uses inverse Gaussian regression with the beta transformation. All these methods are parametric, requiring functional form and distribution assumptions. Both academic studies (for instance, Bastos (2010)) and industry practice sometimes use the non-parametric regression tree in LGD analysis. So far no academic study or industry practice has investigated the relevance of the neural network for LGD modeling. Furthermore, there is a lack of a comprehensive study on the relative

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[1] The Basel II risk parameters are probability of default (PD), loss given default (LGD) and exposure at default (EAD). Effective maturity (M) is also needed for corporate, sovereign, and bank exposures.

[2] See Altman et al. (2005) for a comprehensive survey of studies on default recovery rates.

[3] For example, Covitz and Han (2004), Acharya et al., (2007), Caselli et al., (2008), and Qi and Yang (2009).

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performance of all these models. In particular, non-parametric methods are more prone to data over-fitting than parametric methods, which may lead to inferior out-of-sample performance. Given the importance of LGD in credit risk analysis, a good understanding of these methods is crucial for fixed-income investors, rating agencies, bankers, bank regulators, and academics.

This study aims to fill the gap in the understanding of the various methods to model LGD. We employ a large dataset comprising of 3751 defaulted securities in the US from 1985 to 2008 to compare various LGD modeling methods. Our sample includes both bank loans and bonds. We examine a total of six methods, four of which are parametric – ordinary least squares regression (OLS), fractional response regression (FRR), inverse Gaussian regression (IGR), and inverse Gaussian regression with beta transformation (IGR-BT), and the other two are non-parametric – regression tree (RT) and neural network (NN).

We find that the non-parametric methods provide better fit and more accurate prediction than the parametric methods. Among the parametric methods, although fractional response regression has a slight edge over OLS, both methods perform reasonably well. Performance of the two transformation regression methods (i.e., inverse Gaussian and inverse Gaussian with beta transformation) is very sensitive to the choice of ε, a small value added to LGD of 0 and subtracted from LGD of 1 prior to transformation. Very small or very large ε's can result in inferior model fit. Our findings thus suggest that these transformation methods need to be used with caution. Although fitted LGDs from all methods show some degree of bi-modal distributions, the two transformation methods with extremely small ε's generate bi-modal patterns with heavy concentrations near LGD of 0 and 1, even though these models yield poor model fit. This evidence suggests that a model's ability to produce a strong bi-modal distribution does not necessarily lead to accurate LGD estimation.

All models, including the non-parametric methods when the over-fitting problem is under control, are able to generate quite stable out-of-sample results. However, inferences on some instrument-level explanatory variables are slightly different between models, suggesting that these instrument-level variables may not have a clear-cut linear relation with LGD. This finding implies that the current industry practice, which relies solely on instrument-level variables to predict LGDs, may not be adequate. Additional variables and non-linear or non-parametric relations should also be considered.

The rest of paper proceeds as follows. In the next section, we discuss different methods, and Section 3 describes the data. We present results in Section 4, discuss robustness tests in Section 5, and the final section concludes.

2. Estimation methods

2.1. Fractional response regression (FRR)

This simple quasi-likelihood method was proposed by Papke and Wooldridge (1996) to model a continuous variable ranging between 0 and 1 and to perform asymptotically valid inference. The model specification is as follows:

$$E(\mathrm{LGD}|x) = G(x\beta), \tag{1}$$

where x is a vector of explanatory variables, β is a vector of model parameters, and the functional form for $G(\cdot)$ is usually the logistic function,

$$G(x\beta) = \frac{1}{1+\exp(-x\beta)}, \tag{2}$$

or the log-log function,

$$G(x\beta) = \exp(-\exp(-x\beta)). \tag{3}$$

Since it is clear that $0 < G(z) < 1$ for all $z \in R$, the predicted value from fractional response regression is bounded between 0 and 1.

To estimate the coefficients β, we maximize the following log-likelihood function:

$$\sum_i l_i(\hat{\beta}) = \sum_i \left\{ \mathrm{LGD}_i \times \log\left[G(x_i\hat{\beta})\right] + (1-\mathrm{LGD}_i) \times \log\left[1-G(x_i\hat{\beta})\right] \right\}. \tag{4}$$

FRR does not need to use *ad hoc* transformations to handle observed LGDs of 0 or 1, and the estimators of β from FRR are consistent and asymptotically normal.

2.2. The transformation regressions

Because LGDs are bounded in the unit interval [0, 1], whereas the predicted LGDs from an OLS regression are not bounded, certain transformation can be applied to LGDs before running the regression and the fitted LGDs from the regression are then transformed back to (0, 1). We consider two commonly used transformation regressions – inverse Gaussian that addresses the [0, 1] boundary, and beta transformation that considers the bi-modal LGD distribution, in addition to the [0, 1] boundary.

2.2.1. Inverse Gaussian regression (IGR)

First, we transform LGDs from the unit interval $(0, 1)$ to $(-\infty, \infty)$ using an inverse Gaussian distribution function. We then run an OLS regression using the transformed LGDs, and finally, we transform the fitted values back from $(-\infty, \infty)$ to $(0, 1)$ using the Gaussian distribution function. We choose the Gaussian distribution function out of convenience.

2.2.2. Inverse Gaussian regression with beta transformation (IGR-BT)

In this approach, we assume that LGDs follow a beta distribution. First, we use the realized LGDs to estimate the beta distribution parameters α and β. The cumulative probabilities are then calculated using these estimated beta distribution parameters and transformed from $(0, 1)$ to $(-\infty, \infty)$ using an inverse Gaussian distribution. We then run an OLS regression and transform the fitted values from OLS back from $(-\infty, \infty)$ to $(0, 1)$ using the Gaussian distribution. We finally convert the probabilities back from uni-modal Gaussian distribution to bi-modal beta distribution using the inverse beta distribution. So this method is quite similar to the inverse Gaussian regression, except that it assumes a beta distribution for LGDs and uses the beta distribution to pre-process and post-process LGDs.

2.2.3. Treatment for LGD values of 0 and 1

Neither inverse Gaussian nor beta transformation is defined when LGD equals 0 or 1, and thus, we need to adjust LGD values of 0 or 1 by adding or subtracting a small positive value ε. To minimize LGD data distortion, a natural choice for ε would be a very small number. However, if ε is very small, the transformed value of LGD = 0 (or ε) is a finite number very close to $-\infty$, while the transformed value of LGD = 1 (or $1-\varepsilon$) is a finite number very close to ∞. If a substantial proportion of the observations have LGDs equal to 0 or 1, the transformed values will have very large spikes near $-\infty$ or ∞, which are likely to result in poor model fit and inaccurate predictions. On the other hand, larger ε could lead to larger differences between the actual LGDs and the LGDs used for model estimation when LGDs equal 0 or 1, which is likely to result in poor model fit and inaccurate LGD predictions, too. Therefore, it is important to investigate model sensitivity to different ε's and whether there exists an optimal ε that balances the impact of data distortion and extreme values on model performance.

2.3. Regression tree (RT)

Regression tree is popularized by Breiman et al. (1984) and can be used to predict continuous dependent variables or categorical predictor variables. We derive predictions from a few simple if-then conditions and partition the dataset recursively into smaller mutually exclusive subsets.

The split algorithm of a regression tree minimizes intra-subset variance. It starts with a root node that contains all observations. It then searches over all possible binary splits of all variables to generate the smallest intra-subset variation of the LGD in the next level of nodes. The predicted LGDs are the average value of the LGDs in each leaf node. This process is repeated recursively until no further reduction in variance is possible or certain stopping criterion is met. Because LGDs are in the range of [0, 1] and predictions are given by averages of LGDs in each leaf node, predicted LGDs from a regression tree will be bounded in [0, 1].

Results from a regression tree are easy to interpret and communicate, which makes it very appealing to users. In addition, regression trees can easily handle non-linearity because they approximate non-linear functions by piece-wise constant ones. Further, this method is fairly robust to outliers.

Although over-fitting can be a problem for all methods, it is more of a concern for non-parametric methods, including regression tree. Given the noise in any real data, determining when to stop splitting the tree is important when building a regression tree. For example, 9 splits applied to a dataset of only 10 observations would result in perfect fit in sample, but the resulting model is unlikely to generate robust prediction out of sample. In general, near perfect fit can be achieved with a sufficient number of splits, but an overly complex tree cannot produce accurate out-of-sample prediction. One way to alleviate the concern of over-fitting is to floor the number of observations in each leaf node. Further, it is important to conduct cross-validation tests by applying the tree built from one set of observations (development sample) to another completely independent set of observations (validation sample). If most of the splits in the development sample are driven by noise, then the prediction on the validation sample would be poor.

We use 10-fold cross-validation to examine the usefulness of a regression tree. We divide the entire sample randomly into 10 mutually exclusive subsets of roughly the same size and reserve each of the 10 subsets as the validation sample with the model estimated using the remaining 9 subsets. We report the sum of SSE from each of the 10 validation subsets and *R*-squared based on this sum of SSE. To be consistent, we also apply the same 10-fold cross-validation to other modeling methods as well.

2.4. Neural network (NN)

Neural networks are a class of flexible non-linear models inspired by the way human brain processes information. Given an appropriate number of hidden-layer units, neural networks can approximate a non-linear (or linear) function to an arbitrary degree of accuracy through the composition of a network of relatively simple functions (see White (1990)). Among various types of neural networks, the three-layer feedforward network is the most widely used. Let f be the unknown underlying function (linear or non-linear), through which a vector of input variables x explain *LGD*, i.e., $LGD = f(x)$. We can then approximate f with a three-layer neural network model:

$$f(x) = \alpha_0 + \sum_{j=1}^{n} \alpha_j G\left(\sum_{i=1}^{k} \beta_{ij} x_i + \beta_{0j}\right) + \varepsilon, \tag{5}$$

where n is the number of units in the hidden layer, k is the number of x variables; G is the logistic function as in Eq. (2), a commonly used transfer function in feedforward neural networks; $\{\alpha_j, j = 0, 1, \ldots, n\}$ represents a vector of coefficients from the hidden-layer units to the output-layer units; $\{\beta_j, i = 0, 1, \ldots, k, j = 0, 1, \ldots, n\}$ denotes a matrix of coefficients from the input-layer units to the hidden-layer units; and ε is the error term. The error term can be made arbitrarily small if n is sufficiently large. However, too large an n can cause the model to overfit in which case the in-sample errors are small but the out-of-sample errors may be large. The choice of n is data dependent and there exists no general rule for predetermining it. Thus, we perform sensitivity analysis by exploring different values of n, ranging from 5, 10, 15, and so on, up to 40.

We estimate the parameters by minimizing the sum of squared errors $\Sigma\varepsilon^2$. We use the Levenberg–Marquardt algorithm, as it is by far the fastest algorithm for moderate-sized (up to several hundred free parameters) feedforward neural networks. We generate the initial values of the parameters with Nguyen and Widrow's (1990) method and use early stopping improve NN's out-of-sample performance.[4] We also rely on 10-fold cross-validation to examine the stability of out-of-sample performance.

3. Sample

Our sample is from the Moody's Ultimate Recovery Database (URD). The data coverage is US corporate default events with over $50 million in debt at the time of default. It has three alternative approaches of calculating recovery: the settlement method, the trading price method, and the liquidity event method.[5] The database also provides Moody's preferred method, which varies for each default and is the one Moody's consider the most representative of the actual recovery. The most common preferred method is the settlement method. We use the LGD numbers from the preferred method for all our analysis. To obtain discounted ultimate recoveries, each nominal recovery in URD is discounted back to the last time when interest was paid using the instrument's pre-petition coupon rate. In total, we have 3751 observations from 1985 to 2008.

Panel A in Table 1 shows the number of LGD observations, mean, median, and standard deviation by year. There are more observations in years 2001 and 2002, coinciding with the high default rates in these 2 years. The mean LGD ranges from a low of 19.16% in 2007 to a high of 85.61% in 1985. The median LGD reaches a low of 0 in 2006–2007. As is shown in the last row of the panel, the standard deviation of the mean LGD by year is 15.19%, while that of the median LGD by year is 25.29%; these numbers are low relative to the overall mean and median at 45.29% and 44.23%, respectively. However, the apparently low variance of LGDs from year to year does not necessarily suggest that the LGD is highly predictable for individual bonds or loans. The standard deviation of LGDs in each year is in the range of 30–40%, suggesting that there is a wide range of variations in LGDs across individual instruments. The same conclusion can be found in Fig. 1, which shows that LGDs in our sample are quite widely spread out, with a heavy concentration at both ends in the intervals of [0, 0.1) and [0.9, 1]. This bi-modal distribution implies that upon default, the most likely outcome is losing almost nothing (over 30% chance) or losing almost everything (over 15% chance).

[4] The entire sample is randomly divided into training, validation and test sets. The training and validation errors normally decrease during the initial phase of training then start to rise when the network begins to overfit the data. The validation error is monitored during training, and the training is stopped when the validation error increases for a specific number of iterations. The coefficients at the minimum of the validation error are returned.

[5] With the settle method the value of the settlement instruments is taken at or close to emergence. With the trading price method the value is based on the trading price of the defaulted instrument taken at or post-emergence. With the liquidity method the value of the settlement instruments is taken at the time of a liquidity event, such as the maturity of the instrument, the call of the instrument, or a subsequent default event.

Table 1
LGD by year, industry, instrument type, and collateral type.

	Number of observations	Median (%)	LGD mean (%)	Std (%)
Panel A. By year				
1985	2	85.61	85.61	0.33
1987	39	33.04	37.50	31.68
1988	54	61.03	79.47	36.12
1989	72	61.64	69.18	30.69
1990	179	56.50	61.11	37.30
1991	256	40.86	35.30	37.25
1992	103	43.70	41.15	39.95
1993	96	45.29	57.97	39.74
1994	58	35.54	33.06	36.57
1995	110	36.02	30.52	37.22
1996	66	37.42	21.00	38.95
1997	60	37.47	20.30	39.81
1998	73	58.86	64.20	34.43
1999	180	45.74	39.76	39.15
2000	281	49.64	54.32	39.12
2001	573	52.25	64.21	40.36
2002	688	55.17	68.18	34.69
2003	335	33.11	26.00	36.24
2004	175	25.71	1.75	33.90
2005	182	25.38	12.55	30.63
2006	68	26.60	0.00	35.75
2007	35	19.16	0.00	29.00
2008	66	52.26	52.77	37.12
Overall	3751	45.29	44.23	38.34
Std		15.19	25.29	
Panel B. By industry				
Consumer non-durable goods	242	40.03	38.40	34.77
Consumer durables	157	41.72	48.15	36.34
Manufacturing	433	39.00	32.74	38.35
Energy/Natural resources	198	48.33	57.16	35.94
Chemical	63	29.14	0.00	37.36
Business equipment	238	46.93	45.27	37.97
Telecommunications	552	56.04	68.18	37.79
Utilities	229	16.32	0.00	27.42
Shops	616	49.63	53.61	39.80
Health care	150	50.99	58.42	38.69
Financial institutions	111	49.67	61.69	41.12
Other	762	46.94	45.35	36.87
Panel C. By instrument type				
Revolvers	722	18.32	0.00	28.90
Term loans	662	27.61	3.33E−09	34.02
Senior secured bonds	505	40.08	41.91	32.94
Senior unsecured bonds	1033	54.73	61.13	35.40
Senior subordinated bonds	413	73.18	84.33	29.67
Subordinated bonds	352	74.25	86.08	31.71
Junior bonds	64	81.77	95.60	27.98
Panel D. By collateral type				
Capital Stock	164	34.09	31.77	30.14
Equipment	119	56.13	79.20	34.36
Guarantees	11	0.85	0.00	2.83
Intellectual	5	54.72	62.64	32.62
Inter-company debt	1	87.35	87.35	
Inventory, receivables and cash	144	6.24	0.00	19.77
Most assets	1198	23.45	0.00	30.46
Other assets	31	21.40	0.00	31.37
Unsecured	1924	62.62	75.32	35.11
Second lien	126	42.04	41.76	36.16
Third lien	28	64.27	76.34	35.75

Panel B shows the sample distribution and LGDs by each industry. Consistent with earlier studies (e.g., Altman and Kishore (1996) and Acharya et al. (2007)), we find that the utility industry has the lowest LGD, both in terms of mean and median; its LGD standard deviation is also the lowest among all industries. Following the convention, we will include a utility industry dummy to reflect this phenomenon.

Panel C shows the sample breakdown by instrument type. There are seven different debt instrument types: revolvers, term loans, senior secured bonds, senior unsecured bonds, senior subordinated bonds, subordinated bonds, and junior bonds. Senior unsecured bonds constitute the largest proportion of our sample with 1033 observations, followed by revolvers (722 observations) and term loans (662 observations). Junior bonds constitute the lowest share of our sample. The mean LGD is the lowest among revolvers at 18%, and highest among junior bonds. The increase in LGD across instrument types is consistent with the notion of debt seniority. Further, there is relatively less variation in LGD among revolvers and junior bonds. Senior unsecured bonds have the highest variation in LGD.

Panel D provides the sample breakdown by collateral type. We group collaterals into 11 types: capital stock; equipment; guarantee; intellectual property; inter-company debt; inventory, receivables, and cash; "most assets"; other assets; unsecured; second lien; and third lien.[6] The majority of our sample debt instruments are unsecured, and the LGDs from these instruments are quite high, with a mean at 62.62%, and median at 75.32%.

A small number of the defaulted instruments of certain firms in our sample are guaranteed by a separately rated entity. For example, Motorola provides guarantee to debt instruments of Iridium. Panel D shows that when default happens, these guaranteed debt instruments have the lowest LGD (mean at 0.85%), with the lowest variation (standard deviation at 2.83%). Inventory, receivables, and cash also provide good collateral values – defaulted instruments with this type of collateral have a mean LGD at 6.24% and median LGD at 0. Defaulted instruments secured by capital stock have a higher LGD than those secured by inventory, receivables, and cash. This is not surprising because if a large proportion of the collateral is the stock of the company, they should not worth much when the firm is in default. LGDs of defaulted exposures secured by "most assets" and other assets are also quite low, with means at 23.45% and 21.40%, respectively.

The LGDs from debt instruments secured by equipment are quite high, with a mean at 56.13% and median at 79.20%. A possible explanation for this phenomenon is that equipment is industry- or firm-specific, which may not command a high value in a fire sale. Debts secured by intellectual assets have high LGDs; this result is intuitive because the intangible assets may be of only specific value to the default firm. There is only one case of defaulted exposure secured by inter-company debt and it carries a very high LGD of 87.35%.

Finally, the LGDs of second liens have a mean of 42.04% and a median of 41.76%; they are lower than the mean of 45.25% and the median of 43.93% for the first liens. Third liens have a mean LGD of 64.27% and a median of 76.34%; these numbers are comparable to those of unsecured. These results suggest that the second- and third-lien holders may not fare the worst when default occurs.

Table 2 provides summary statistics of the explanatory variables used in this study. Distance-to-default is a measure of volatility-adjusted leverage, and it is backed out of the Merton (1974) model.[7] We use 12 industries as defined in Panel B of Table 1. The mean industry distance-to-default is 15.00 and the median

[6] Most assets (all assets excluding inventory and account receivables), all assets, PP&E and all non-current assets are grouped together as "most assets". Equipment is kept separate because exposures secured by equipment have much higher LGDs than those secured by all the collateral types grouped into "most assets." We group oil and gas properties and real estate into other assets due to low numbers of observations and the minimal difference between the LGDs of these two collateral types. Guarantees, intellectual and inter-company debt are kept separate despite the low numbers of observations, because these collateral types are quite unusual and are associated with LGDs distinctively different from other collateral types. We classify cash in the same group as inventory and receivables because they are all liquid assets and have similar LGDs. On the other hand, capital stock, to which cash is often grouped with, has fairly high LGDs.

[7] Monthly volatility is the volatility of daily returns during the month. We use Compustat quarterly leverage data. We would like to thank Shumway for providing the SAS codes.

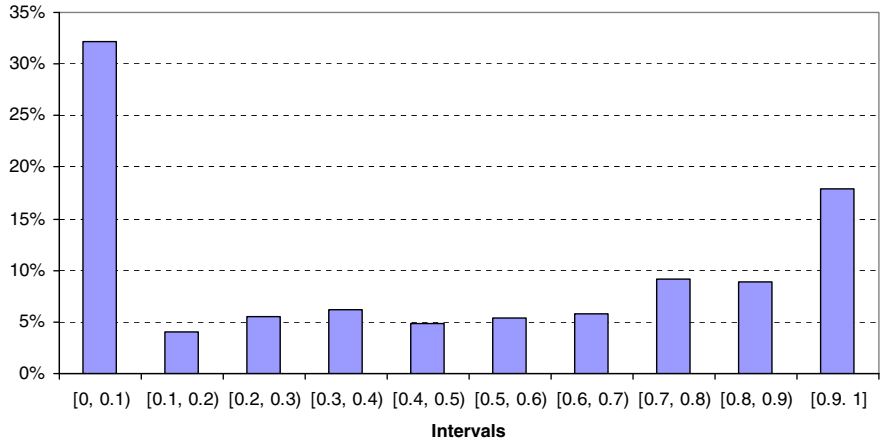


Fig. 1. Distribution of observed LGDs.

Table 2
Summary statistics.

	Mean	Median	Std
Panel A. Summary statistics			
Industry distance-to-default (%)	15.00	13.65	5.73
Trailing 12-month industry default rate (%)	3.10	2.05	3.53
Aggregate distance-to-default (%)	17.13	15.92	4.22
Trailing 12-month aggregate default rate (%)	1.91	1.89	1.01
Trailing 12-month market return (%)	3.19	5.95	17.90
3-month T-bill rate (%)	3.63	3.51	2.02
Percentage above (%)	22.16	2.98	28.95
Seniority index (%)	48.92	50.00	25.14

	LGD	Percentage above
Panel B. Correlations		
Percentage above	0.460	
Seniority index	0.571	0.853

is 13.65; these numbers are lower than those at the aggregate level. The mean trailing 12-month industry default rate is 3.10%, higher than the mean aggregate default rate (1.91%). These results suggest that the sample firms are from troubled industries. We use the NYSE-NASDAQ-AMEX value-weighted index as the market return, and the mean trailing 12-month market return is 3.19%, consistent with the by-year distribution reported in Panel A of Table 1.

The variable percentage above measures the percentage of debt obligations of a firm that is more senior to a particular instrument. This variable is included in Moody's LossCalc. The variable seniority index is a variable we constructed using percentage above plus one-half of the percentage at the same rank. Both percentages are available in URD. The construction of seniority index is motivated by the findings in Bris et al. (2009) that many issuers only issue bonds at the same seniority class, suggesting that only incorporating percentage above may not be sufficient, as each credit holder is likely to recover less in the event of default, the larger the number of instruments held by different creditors are at the same rank. Qi and Zhao (2010) find that this variable is the most important explanatory variable of the recovery rate (or 1-LGD). In our sample, the mean for percentage above is lower than half of the mean for seniority index, suggesting a significant portion of the debt instruments of the same obligor are at the same rank. Panel B of Table 2 shows the correlation between these two variables and LGD. We find that LGD has a lower correlation with percentage above than with the seniority index. The correlation between percentage above and seniority index is 0.853, high but not close to 1, thus there is additional information contained in seniority index not already captured by percentage above, which may be used to improve the fit of an LGD model.

Table 3
Regression results from OLS and fractional response (FRR).

Dependent variable: LGD	OLS		FRR	
Independent variable	Coefficient	*p*-Value	Coefficient	*p*-Value
Seniority index	0.562	<0.001	3.031	<0.001
Revolvers	−0.119	<0.001	−0.477	<0.001
Term loans	−0.075	0.012	−0.145	0.013
Senior secured bonds	−0.01	0.755	0.116	0.074
Senior unsecured bonds	−0.079	<0.001	−0.335	<0.001
Senior subordinated bonds	0.032	0.128	0.175	<0.001
Junior bonds	0.020	0.608	0.181	0.180
Capital stock	0.006	0.810	0.056	0.141
Equipment	0.104	0.001	0.440	<0.001
Guarantees	−0.259	0.003	−2.388	0.438
Intellectual	0.274	0.034	1.448	0.171
Inter company debt	0.443	0.121	1.144	0.811
Inventory, receivables, and cash	−0.096	<0.001	−1.241	<0.001
Other	−0.065	0.211	−0.272	0.195
Unsecured	0.145	<0.001	0.717	<0.001
Second lien	0.065	0.020	0.295	<0.001
Third lien	0.192	0.001	1.036	<0.001
Industry distance-to-default	−0.953	<0.001	−5.724	<0.001
Aggregate default rate	0.311	<0.001	1.586	<0.001
Trailing 12-month market return	−0.011	<0.001	−0.062	<0.001
3-month T-bill rate	0.128	<0.001	0.768	<0.001
Utility dummy	−0.197	<0.001	−1.011	<0.001
Intercept	0.211	<0.001	−1.588	<0.001
Observations	3751		3751	
R-squared	0.448		0.463	
SSE	304.32		296.12	
10-fold cross-validation				
R-squared (std)	0.443	(0.005)	0.457	(0.047)
SSE (std)	306.92	(7.53)	298.60	(7.87)

4. Results

This section compares results from different modeling methods for LGD. We use the same set of independent variables across all methods except for the regression tree, for which we include three additional variables (percentage above, industry default rate, and aggregate distance-to-default). This is because all methods except for the regression tree could suffer from the multi-collinearity problem as these three variables are highly correlated with the other explanatory variables.[8] Also in all but the regression tree methods, we use subordinated bonds as the base instrument and

[8] Our conclusions do not change if we use the same set of independent variables in the regression tree. These results are available upon request.

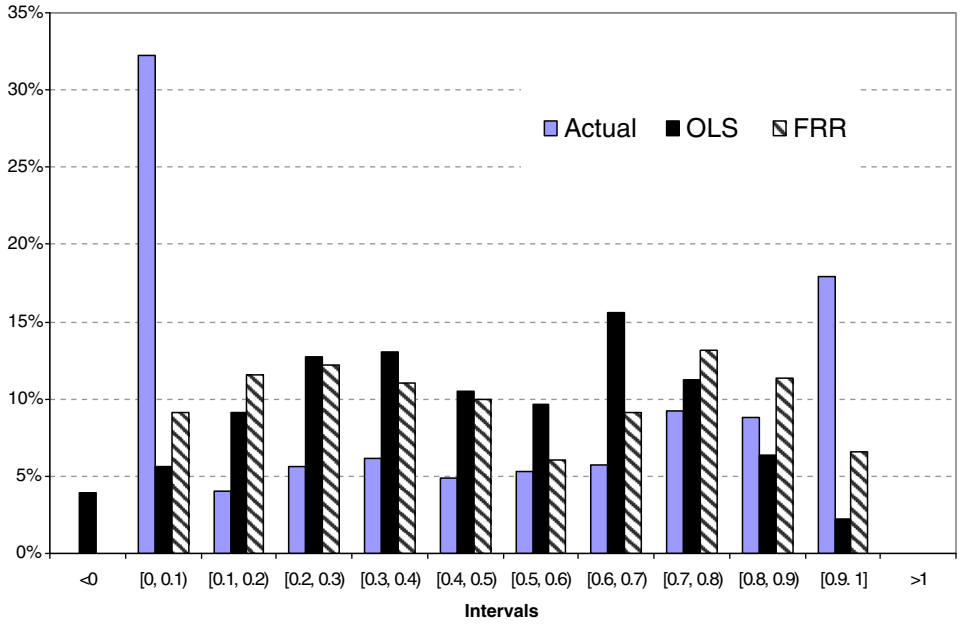


Fig. 2. Distribution of actual versus fitted LGDs from OLS and FRR.

Table 4
Results using different ε in the transformation regressions.

	In sample		10-fold cross-validation			
	R-squared	SSE	*R*-squared	(Std)	SSE	(Std)
Panel A. Inverse Gaussian regression (IGR)						
1.00E−11	0.171	456.93	0.167	(0.089)	459.42	(14.71)
0.0001	0.327	370.84	0.323	(0.078)	373.48	(13.04)
0.0005	0.357	354.24	0.353	(0.075)	356.89	(12.50)
0.001	0.372	346.29	0.367	(0.073)	348.94	(12.21)
0.005	0.408	326.23	0.403	(0.068)	328.88	(11.32)
0.01	0.424	317.39	0.420	(0.065)	320.03	(10.82)
0.05	**0.452**	**302.29**	**0.447**	**(0.056)**	**304.85**	**(9.18)**
0.08	0.451	302.93	0.446	(0.052)	305.43	(8.52)
0.1	0.447	305.15	0.442	(0.050)	307.62	(8.15)
0.2	0.406	327.28	0.402	(0.043)	329.58	(6.80)
0.3	0.345	361.32	0.341	(0.038)	363.44	(5.82)
0.5	0.165	460.34	0.162	(0.033)	462.03	(4.78)
Panel B. Inverse Gaussian regression with Beta transformation (IGR-BT)						
1.00E−11	0.111	490.33	0.106	(0.091)	492.75	(14.94)
0.0001	0.309	380.77	0.305	(0.081)	383.39	(13.42)
0.0005	0.345	361.03	0.340	(0.077)	363.68	(12.81)
0.001	0.362	351.76	0.357	(0.075)	354.40	(12.49)
0.005	0.403	328.92	0.399	(0.069)	331.57	(11.51)
0.01	0.421	319.13	0.416	(0.066)	321.78	(10.97)
0.05	**0.451**	**302.73**	**0.446**	**(0.056)**	**305.30**	**(9.24)**
0.08	0.450	303.15	0.446	(0.052)	305.66	(8.55)
0.1	0.446	305.26	0.442	(0.050)	307.74	(8.18)
0.2	0.407	326.87	0.403	(0.043)	329.17	(6.79)
0.3	0.346	360.43	0.342	(0.037)	362.55	(5.78)
0.5	0.168	458.65	0.165	(0.032)	460.33	(4.71)

Bold numbers indicate the optimal model specification that has the highest *R*-squared and the lowest SSE among the alternative specifications listed in the table.

"most assets" as the base collateral type. We report regression results from OLS and fractional response in Table 3 and Fig. 2, results from the inverse Gaussian, and inverse Gaussian with beta transformation in Tables 4 and 5 and Figs. 3.1 and 3.2, and results from regression tree and neural network in Tables 6 and 7 and Figs. 4–7.

4.1. Ordinary least squares regression (OLS)

First of all, the coefficient to the new variable seniority index is 0.562 and is highly significant. Therefore, a 10% increase in the percentage of debt more senior than or equal to the instrument is likely to lead to 5.6% higher LGD. Further, OLS results suggest that revolvers and term loans are associated with significantly lower LGDs, and on average, revolvers' LGD is about 4% points lower than the LGD of term loans, after controlling for other factors. LGDs of senior secured bonds are not significantly different from those of subordinated bonds,[9] while LGDs of senior unsecured bonds are sig-

[9] This result is quite surprising and is not consistent with conventional wisdom and the summary statistics in Panel C of Table 1. We find that this result is driven by the inclusion of the utility industry dummy – excluding the utility dummy, the coefficient of the senior secured bonds is significantly negative.

Table 5
Regression results from inverse Gaussian (IGR) and inverse Gaussian with beta transformation (IGR-BT) ($\varepsilon = 0.05$).

Dependent variable: LGD	IGR		IGR-BT	
Independent variable	Coefficient	*p*-Value	Coefficient	*p*-Value
Seniority index	1.962	<0.001	1.453	<0.001
Revolvers	−0.401	<0.001	−0.297	<0.001
Term loans	−0.252	<0.001	−0.187	<0.001
Senior secured bonds	−0.030	0.588	−0.024	0.562
Senior unsecured bonds	−0.250	<0.001	−0.186	<0.001
Senior subordinated bonds	0.165	<0.001	0.123	<0.001
Junior bonds	0.101	0.139	0.075	0.135
Capital stock	0.012	0.779	0.008	0.799
Equipment	0.231	<0.001	0.167	<0.001
Guarantees	−0.849	<0.001	−0.626	<0.001
Intellectual	0.815	<0.001	0.599	<0.001
Inter company debt	1.364	0.006	1.006	0.007
Inventory, receivables, and cash	−0.292	<0.001	−0.214	<0.001
Other	−0.248	0.007	−0.184	0.007
Unsecured	0.492	<0.001	0.364	<0.001
Second lien	0.197	<0.001	0.145	<0.001
Third lien	0.610	<0.001	0.450	<0.001
Industry distance-to-default	−3.210	<0.001	−2.376	<0.001
Aggregate default rate	1.058	<0.001	0.780	<0.001
Trailing 12-month market return	−0.040	<0.001	−0.029	<0.001
3-month T-bill rate	0.476	<0.001	0.352	<0.001
Utility dummy	−0.687	<0.001	−0.507	<0.001
Intercept	−0.997	<0.001	−0.658	<0.001
Observations	3751		3751	
R-squared	0.452		0.451	
SSE	302.29		302.73	
10-fold cross-validation				
R-squared (std)	0.447	(0.056)	0.446	(0.056)
SSE (std)	304.85	(9.18)	305.30	(9.24)

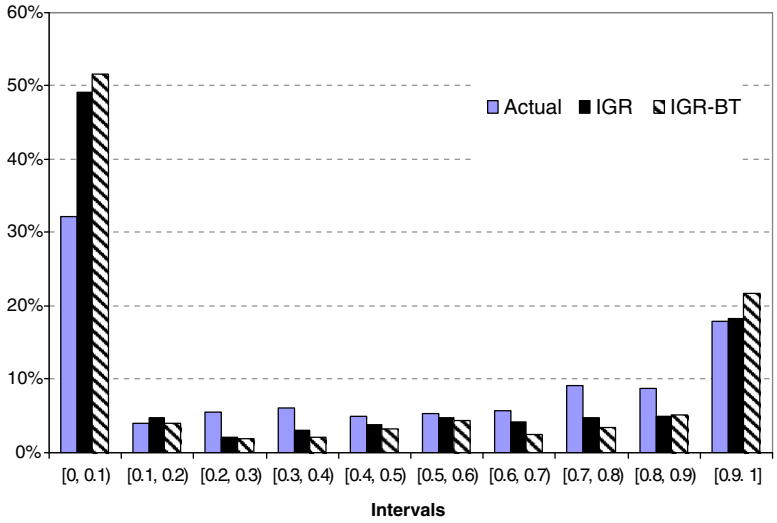


Fig. 3.1. Distribution of fitted LGDs from IGR and IGR-BT ($\varepsilon = 1.00\text{E}{-11}$).

nificantly lower. There is not much difference in LGDs across senior subordinated bonds, subordinated bonds, and junior bonds. On the other hand, unsecured debts, debts backed by equipment, intellectual properties, second and third liens tend to have higher LGDs than instrument backed by "most assets" (the base collateral), while exposures backed by guarantees, inventory, receivables and cash tend to have lower LGDs. Further, LGDs are higher when the aggregate default rates are higher and the 3-month T-bill rate is higher; LGDs are lower when the industry distance-to-default is higher, the trailing 12-month market return is higher, and when the issuer belongs to the utility industry.

The *R*-squared is 0.450 and the SSE is 304.32 from the overall regression. They are 0.448 and 306.96, respectively, from the 10-fold cross-validation, with very low standard deviations. These statistics indicate that the OLS model is fairly stable.

We plot the OLS-fitted LGDs in Fig. 2. Although some fitted LGDs are outside the range of $[0, 1]$, these numbers constitute a very small proportion (around 4%) of the sample. The distribution of the fitted LGDs is slightly bi-modal, however, the higher concentrations are found in the intervals of $[0.3, 0.4)$ and $[0.6, 0.7)$, instead of $[0, 0.1]$ and $[0.9, 1]$, respectively, for the actual LGDs.

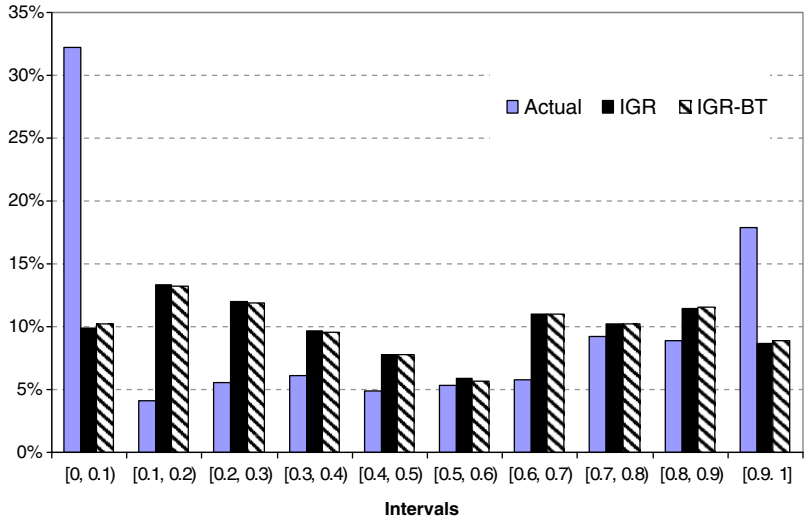


Fig. 3.2. Distribution of fitted LGDs from IGR and IGR-BT ($\varepsilon = 0.05$).

4.2. Fractional response regression (FRR)

Results from FRR are qualitatively similar to those from the OLS.[10] The exceptions are (1) senior secured bonds and the senior subordinated bonds seem to have significantly higher LGDs than the base instrument (subordinated bonds),[11] and (2) the collateral types of guarantees and intellectuals are not significantly related to LGD. Similar to OLS, the 10-fold cross-validation *R*-squared and SSE from FRR are quite close to those in sample with low standard deviations, suggesting that the underlying relationship is rather stable. Further, both the in-sample and out-of-sample *R*-squared are higher and the SSEs are lower for FRR than for OLS. These results indicate that FRR yields better model fit than OLS.

We also plot the histogram of FRR-fitted LGDs in Fig. 2. It is clear that FRR-fitted LGDs do a better job than OLS-fitted LGDs in mimicking the actual LGD distribution. We find high concentrations in the intervals [0.2, 0.3) and [0.7, 0.8). Not surprisingly, all fitted values from FRR fall within [0, 1].

4.3. Transformation regressions (IGR and IGR-BT)

We first investigate in Table 4 results using different ε's. This table shows that results from the transformation regressions are very sensitive to the choice of ε. For both IGR and IGR-BT, the model fit improves initially as ε increases from 1.0E−11 to 0.05 and then deteriorates afterwards; the latter part is not surprising as a higher ε creates larger differences between the actual LGDs and the LGDs used in model estimation. With $\varepsilon = 1E{-}11$, IGR yields an *R*-squared of 0.171 and IGR-BT leads to an *R*-squared of 0.111. The numbers are drastically lower than those of OLS and FRR reported in Table 3. SSEs are also much higher among the transformation methods. The highest *R*-squared and lowest SSE for both methods (both in-sample and 10-fold cross-validation) is achieved at $\varepsilon = 0.05$, suggesting that it may be the optimal epsilon for this sample. At $\varepsilon = 0.05$, the model fit from IGR and IGR-BT matches that of OLS and is slightly worse than that of FRR. Note that the standard deviations of *R*-squared and SSE from the 10-fold cross-validation are very high at very low ε's for both transformation methods. This result suggests that extreme values generated by very small ε's make the transformation methods unstable.

Table 5 reports the detailed results from the transformation methods under $\varepsilon = 0.05$. Results from the two transformation methods are quite similar. The main differences in coefficient estimates of the explanatory variables between these methods and those from either OLS or FRR relate to the instrument type senior subordinated bonds and collateral types guarantees, intellectual, inter-company debt, and other. These variables are non-significant under either OLS or FRR and become significant in Table 5 under both IGR and IGR-BT. The finding that inferences on some instrument-level variables are slightly different between models implies that the instrument-level variables may not have a clear-cut linear relation with LGD. Therefore, the current industry practice, which relies solely on instrument-level variables to predict LGDs, may not be adequate.

We illustrate the distribution of fitted LGDs under $\varepsilon = 1.0E{-}11$ and $\varepsilon = 0.05$ in Figs. 3.1 and 3.2, respectively. The distribution in Fig. 3.1 is strongly bi-modal, with large spikes at both ends. In fact, this distribution shows a stronger bi-modal pattern than the actual LGD distribution. In contrast, the distribution in Fig. 3.2 shows a rather weak bi-modal pattern, quite similar to that of FRR in Fig. 2. Since the model fit associated with Fig. 3.1 is rather poor, these figures suggest that models that can generate a strong bi-modal pattern in fitted LGDs do not necessary lead to good model fit or accurate prediction.

We also study the scatter plots of the actual versus the fitted LGDs from both transformation methods. The fitted LGDs using $\varepsilon = 1.0E{-}11$ do not align well with the actual LGDs, with most of the fitted values deviating substantially away from the 45° line. With $\varepsilon = 0.05$, the fitted LGDs are more clustered along the 45° line, consistent with the higher *R*-squared and lower SSE from $\varepsilon = 0.05$. The scatter plots of actual and fitted LGDs for the transformation methods under $\varepsilon = 0.05$ are quite similar to those from FRR. These scatter plots are not reported due to space limit and are available upon request.

Note that the problem associated with LGDs equal to 0 or 1 is not specific to our sample. The LGDs of 0 or 1 are quite common in wholesale credit exposures due to the absolute priority rules in firms' debt structure. The LGDs of 0 is also common for defaults triggered by distressed exchange. The proportion of 0 LGDs further increases due to the Basel II definition of default – a wholesale obligor is in default if the obligor is deemed by the bank as unlikely to pay its credit obligations in full or is 90 days or more past due on any material credit obligations. Based on this definition, perform-

[10] We only report results using the logistic function – results using the log–log function are quite similar and are not reported due to space limitations.

[11] We find that this result is due to the inclusion of *seniority index*.

Table 6
Results from the regression tree. This table present results from the regression tree method (Breiman et al., 1984). We include in the regression tree all instrument type and collateral type indicators; two industry-level variables: industry distance-to-default (DTD), and trailing 12-month industry default rate; four market-level variables: trailing 12-month aggregate default rate, trailing 12-month stock market returns, aggregate distance-to-default (DTD), and the 3-month T-bill rate; a utility industry dummy, the seniority index, and percentage above.

Minimum size requirement	Maximum steps	In sample		10-fold cross-validation	
		R-squared	SSE	*R*-squared (std)	SSE (std)
Panel A. Regression tree with different minimum size requirements in each leaf					
2	355	0.859	77.65	0.827 (0.269)	95.32 (148.58)
5	342	0.847	84.12	0.804 (0.014)	107.85 (7.90)
10	214	0.767	128.29	0.731 (0.012)	148.22 (6.51)
20	131	0.691	170.19	0.664 (0.018)	185.31 (9.64)
30	87	0.636	200.64	0.616 (0.014)	211.48 (7.70)
40	66	0.61	215.04	0.594 (0.010)	223.93 (5.71)
50	56	0.597	222.12	0.595 (0.038)	228.81 (6.76)
60	46	0.564	240.49	0.55 (0.008)	248.1 (4.11)
70	41	0.545	250.8	0.534 (0.011)	256.74 (6.27)
80	37	0.528	260.44	0.518 (0.007)	265.79 (3.75)
90	30	0.515	267.15	0.506 (0.010)	272.54 (5.15)
100	25	0.501	275.24	0.493 (0.011)	279.36 (6.16)

Split steps	Split variable	Value	In sample		10-fold cross-validation	
			R-squared	SSE	*R*-squared	SSE
Panel B. First 13 splits of the regression tree with a minimum leaf size requirement of 5 observations						
1	Seniority index	0.511	0.252	412.22	0.251	412.8
2	Seniority Index	0.295	0.327	370.76	0.326	371.44
3	Industry distance-to-default	16.585	0.372	346.05	0.371	347.04
4	Industry distance-to-default	14.096	0.397	332.31	0.395	333.32
5	Aggregate distance-to-default	13.83	0.421	319.08	0.419	320.27
6	Industry distance-to-default	8.27	0.434	312.22	0.431	313.56
7	Unsecured	1	0.447	305.14	0.443	306.96
8	3-month T-bill rate	9.40%	0.461	297.38	0.457	299.5
9	Seniority Index	0.213	0.468	293.26	0.464	295.51
10	Industry default rate	4.42%	0.478	287.52	0.474	290.15
11	Aggregate distance-to-default	15	0.489	281.91	0.484	284.55
12	Percentage above	0.77	0.494	278.74	0.49	281.45
13	Seniority index (Std)	0.674	0.499	276	0.495 (0.004)	278.62 (2.05)

ing loans (i.e., LGD = 0) of an obligor that defaults on any of its obligations should be included in the LGD reference data. Therefore, users of the transformation methods should exercise great caution when choosing ε.[12]

4.4. Regression tree (RT)

Table 6 reports results from RT. In Panel A, we show results using different minimum size requirement in each leaf. Panel B presents the first 13 splits of the tree with a minimum of 5 observations at each leaf.

We report in Panel A standard deviations from the 10-fold cross-validation in parenthesis. This panel shows that as the minimum size requirement increases, RT fit worsens, and at a minimum of 5 observations in each leaf, the in-sample *R*-squared is as high as 0.847. Although the difference in *R*-squared and SSE between in-sample and 10-fold cross-validation widens when the minimum size requirement declines, the out-of-sample prediction is still very high at 0.804 at a minimum of 5 observations, with low standard deviations from 10-fold cross-validation. However, the standard deviations explode at a minimum of 2 observations required at each leaf.[13] This shows that RT can indeed suffer from the over-fitting problem, although the problem can be monitored and perhaps avoided by cross-validation. We caution that the seem-

[12] The findings and conclusions contained in this section are based on the two transformation methods we investigated, and do not necessarily have general implications for other transformation methods. Further, the optimal value of $\varepsilon = 0.05$ may be specific to the data used in this study, and thus may not be generalized to other data, even if the same transformation methods are used.

[13] Further, we find that, when the minimum size requirement drops to one observation at each leaf, *R*-squared and SSE from the 10-fold cross-validation worsen.

Table 7
Results from the neural network. This table present results from the neural network method. We used a three-layer feed forward neural network with the number of hidden-layer units ranging from 5 to 40. Early stopping is used to control over-fitting. The entire sample is randomly divided into training, validation and test sets. The optimal number of hidden-layer nodes is chosen by model performance on the validation and testing sets. We include in the model all instrument type and collateral type indicators; one industry-level variable: industry distance-to-default (DTD); three market-level variables: trailing 12-month aggregate default rate, trailing 12-month stock market returns, and the 3-month T-bill rate; a utility industry dummy, and the seniority index.

Hidden-layer nodes	*R*-squared		
	Training	Validation	Testing
5	0.479	0.478	0.384
10	0.587	0.491	0.480
15	0.619	0.498	0.492
20	0.607	0.544	0.481
25	0.624	**0.556**	**0.576**
30	0.662	0.554	0.554
35	0.611	0.517	0.501
40	0.672	0.537	0.497
10-fold cross-validation (Using 25 hidden-layer nodes)			
R-squared (Std)	0.529	(0.031)	
SSE (Std)	259.51	(1.586)	

Bold numbers indicate the optimal model specification that has the highest *R*-squared and the lowest SSE among the alternative specifications listed in the table.

ingly optimal minimum requirement of 5 observations at each leaf may be specific to this sample, thus expect users to investigate the optimal minimum size requirement on their own data. Further, a large tree with 342 splits may not be practical in a production environment.

We report in Panel B the first 13 splits of the tree with the minimum size requirement of 5 observations at each leaf. We show the explanatory variable of the split for each step, the value of split, and the *R*-squared and SSE of the resulting tree. We also report the *R*-squared and SSE from the 10-fold cross-validation. The same steps are also reported in Fig. 4. Because of the large size of this partial tree, we report it in two parts with the left main branch in Fig. 4.1 and the right main branch in Fig. 4.2.

The first split is based on the variable seniority index at the value of 0.511. After the first split, all observations fall into two groups, those with seniority index below 0.511 and those above or equal to 0.511. The predicted LGD is 0.285 for the first group, and 0.674 for the second group; this pattern is qualitatively consistent with the regression results that LGD is positively related to seniority index. This is a very coarse way to predict LGD and not surprisingly, the *R*-squared is relatively low at 0.252, and the SSE is quite high at 412.22. The out-of-sample 10-fold *R*-squared is slightly lower at 0.251 and SSE is slightly higher at 412.80.

The next split is also based on the variable seniority index at the value of 0.295 under the left branch of the first split. The entire sample is now separated into three groups, those with seniority index below 0.295, between 0.295 and 0.511, and those above or equal to 0.511. The fitted LGD for each group is 0.13, 0.41, and 0.67, respectively. With this refinement, the goodness-of-fit is improved: the in-sample *R*-squared climbs to 0.327 and the in-sample SSE drops to 370.76. Those from the 10-fold cross-validation show similar improvement as well.

The third split is based on the variable industry distance-to-default at the value of 16.585 under the right branch of the first split. After this split, the sample is grouped into four segments: (1) those with seniority index below 0.295, (2) those with seniority index between 0.295 and 0.511, (3) those with the seniority index above or equal to 0.511 and industry distance-to-default above or equal to 16.585, and (4) those with the seniority index above or equal to 0.511 and industry distance-to-default below 16.585. The fitted LGD for each of the four groups are: 0.13, 0.41, 0.47, and 0.75. The finding that obligors in industries with a higher distance-to-default tend to have lower LGDs is consistent with the regression results in Tables 3 and 5. After these three splits, the in-sample *R*-squared improves from 0.327 of the previous split to 0.372, and we find similar improvement in the 10-fold cross-validation *R*-squared. The SSE now declines to 346.05 for in-sample fit and 347.04 for out-of-sample fit.

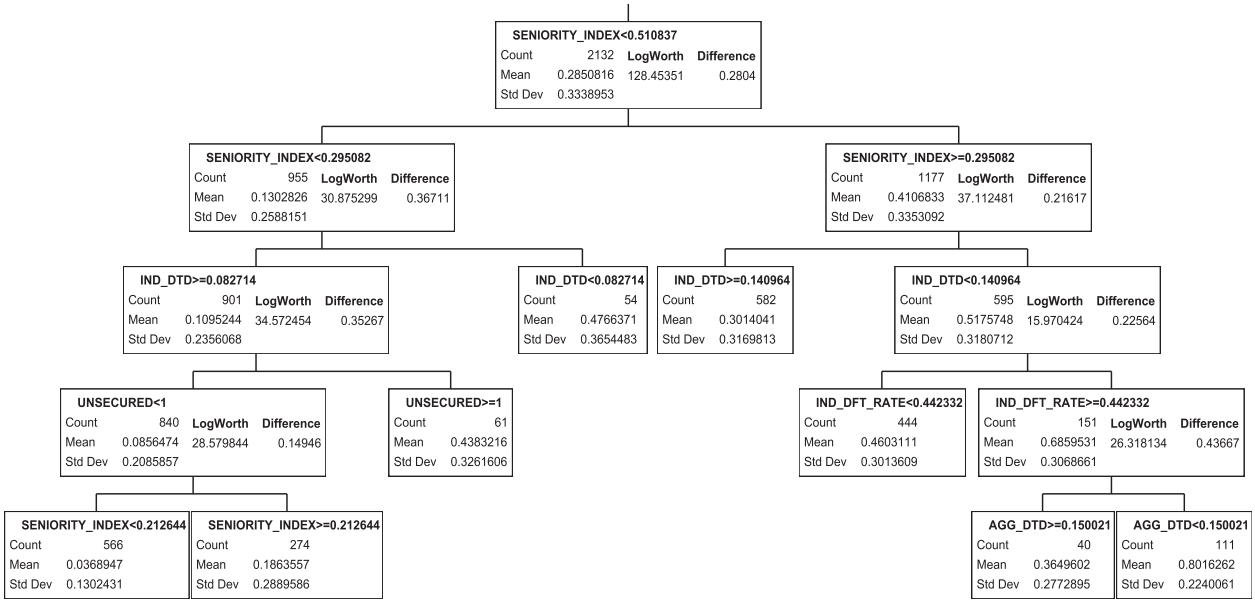


Fig. 4.1. Regression tree. This figure presents the regression tree (Breiman et al., 1984) developed on the small sample. We include in the regression tree all instrument type and collateral type indicators; two industry-level variables: industry distance-to-default (DTD), and trailing 12-month industry default rate; four market-level variables: trailing 12-month aggregate default rate, trailing 12-month stock market returns, aggregate distance-to-default (DTD), and the 3-month T-bill rate; a utility industry dummy, and the *seniority index*. We restrict a minimum of 5 observations in each leaf. Due to space limitation, we only present the first 13 splits with the left main branch shown in Fig. 4.1 and the right main branch in Fig. 4.2. Left main branch.

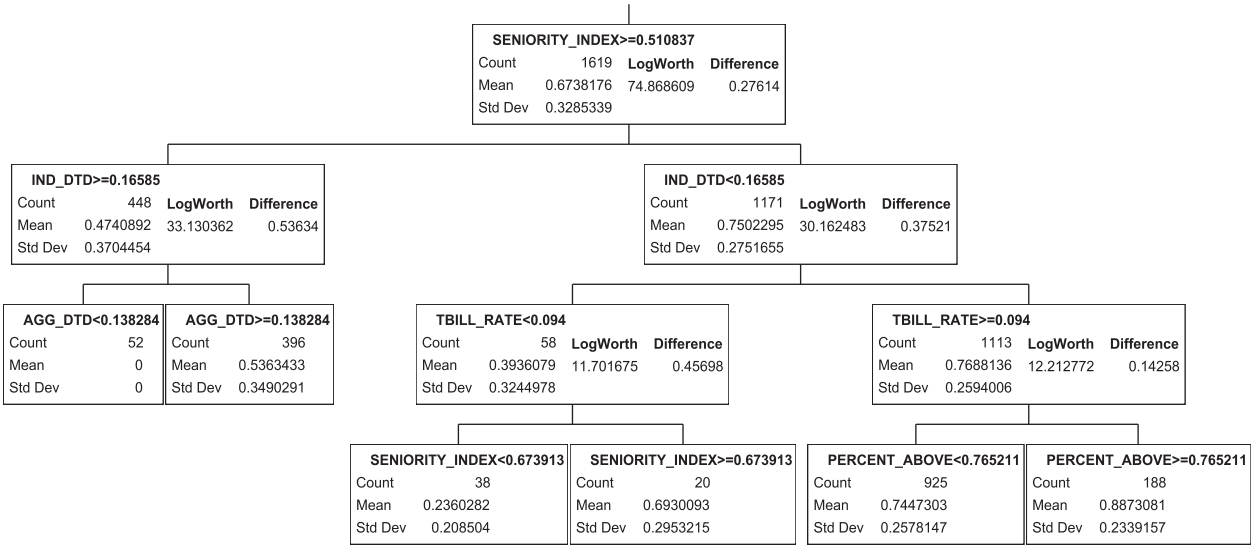


Fig. 4.2. Regression tree. This figure presents the regression tree (Breiman et al., 1984) developed on the small sample. We include in the regression tree all instrument type and collateral type indicators; two industry-level variables: industry distance-to-default (DTD), and trailing 12-month industry default rate; four market-level variables: trailing 12-month aggregate default rate, trailing 12-month stock market returns, aggregate distance-to-default (DTD), and the 3-month T-bill rate; a utility industry dummy, and the *seniority index*. We restrict a minimum of 5 observations in each leaf. Due to space limitation, we only present the first 13 splits with the left main branch shown in Fig. 4.1 and the right main branch in Fig. 4.2. Right main branch.

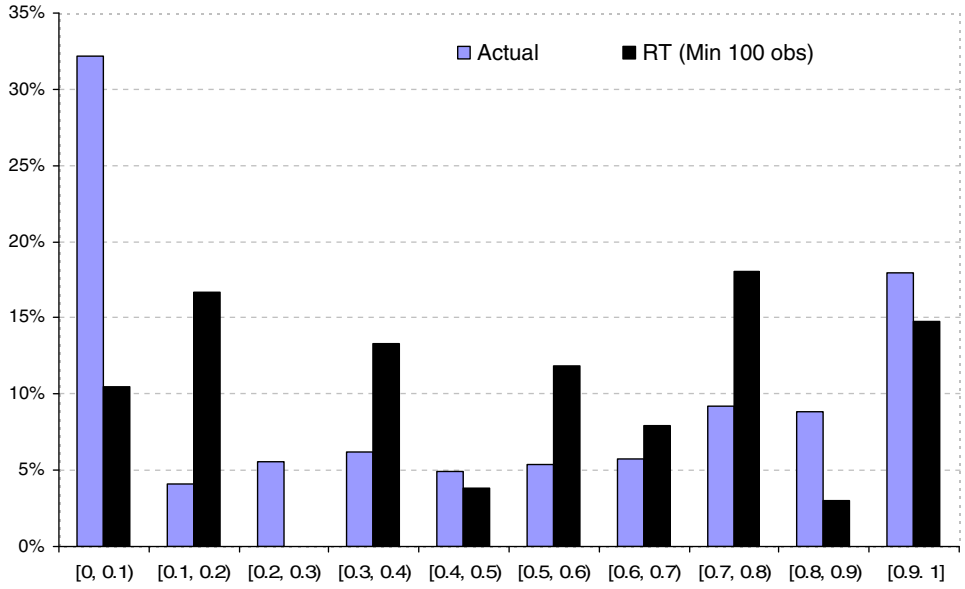


Fig. 5.1. Distribution of fitted LGDs from the regression tree (a minimum size requirement of 100 observations at each leaf).

As the tree grows, both the in-sample and out-of-sample *R*-squared increases, while the SSEs decline, but all at decreasing rates. To achieve the same level of or better goodness-of-fit as OLS or the transformation methods, eight splits are required. To match or beat FRR, we need nine splits. The last row of the table shows that this partial tree is quite stable, with very low standard errors of the *R*-squared and SSE from the 10-fold cross-validation. It therefore seems that in spite of the method's non-parametric nature, when properly controlled, over-fitting does not seem to pose a serious problem for the regression tree method.

Fig. 5.1 reports distribution of fitted LGDs from RT with the minimum leaf size requirement of 100. The highest concentrations are in the intervals [0.1, 0.2) and [0.7, 0.8), rather than [0, 0.1) and [0.9, 1]. Further, there is no strong bi-modal pattern. This finding once more implies that the bi-modal distribution may be of only secondary importance in predicting LGD.

Fig. 5.2 reports distribution of fitted LGDs from RT with the minimum leaf size requirement of 5. The bi-modal distribution pattern in Fig. 5.2 is not as strong as that in Fig. 3.1 – the spike at the interval of [0.9, 1] is not as obvious, although the spike at [0, 0.1) is still large. This finding once again implies that a model's ability to generate strong bi-modal distribution may not necessarily lead to accurate LGD modeling and forecasting. Among all fitted LGD distributions, Fig. 5.2 mimics the distribution of the actual LGDs the best, consistent with the result in Panel A of Table 6 that this tree produces very high predictive accuracy. We caution again that the minimum leaf size of 5 can be too low for other samples, and such a large tree may not be practical.

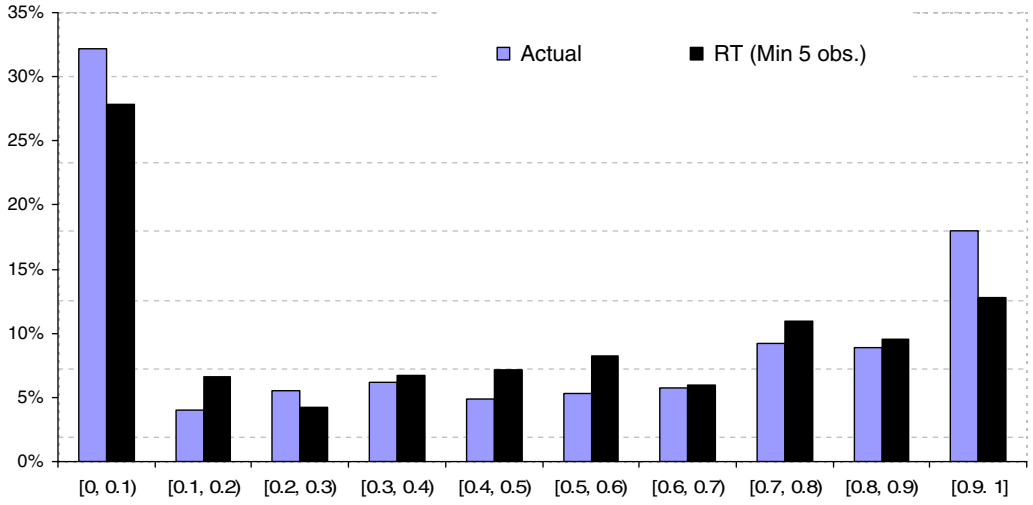


Fig. 5.2. Distribution of fitted LGDs from the regression tree (a minimum size requirement of 5 observations at each leaf).

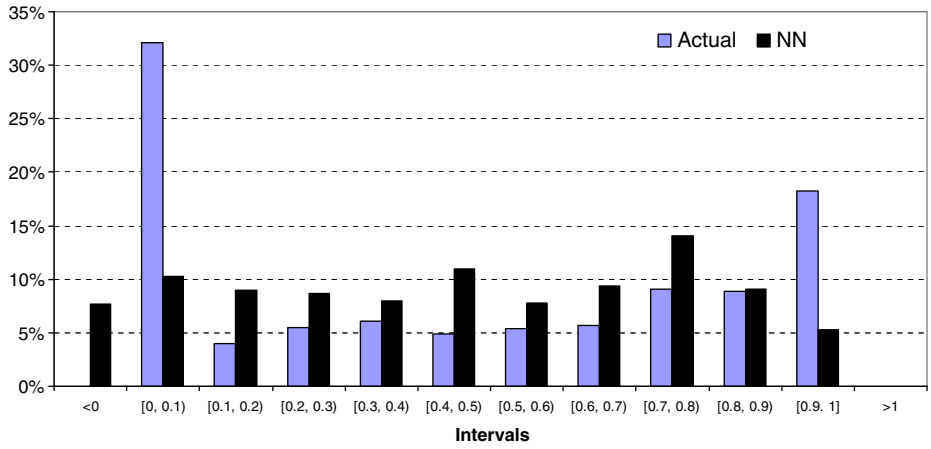


Fig. 6. Distribution of fitted LGDs from the neural network.

4.5. Neural network (NN)

We report results from NN in Table 7. To choose the optimal n, the number of hidden-layer units, we divide the sample randomly into three parts: training, validation and test sets, each representing 70%, 15%, and 15% of the entire sample. The model fit changes when n increases. On the training set, R-squared shows tendency to improve as n becomes larger, whereas on the validation and test sets, R-squared initially improves then deteriorates as n increases, with the peak R-squared of 0.556 and 0.576 on the validation and test sets when n is 25.[14] This suggests that the optimal n for our sample is likely to be 25, thus we use the same n of 25 to generate the 10-fold cross-validation results.

NN also shows good 10-fold cross-validation results. The SSE for the 10-fold cross-validation is the sum of the SSE from each validation sample, and the sum of SSE is then used to compute the R-squared for the 10-fold cross-validation. Table 7 shows that the out-of-sample R-squared of NN is 0.529, which is higher than those of parametric methods, and the out-of-sample SSE of NN is 259.51, which is lower than those of parametric methods. Both the R-squared and SSE have very low standard deviations, indicating the stable performance of NN out of sample. RT with the minimum size requirement at 70 or less shows better model fit than NN in 10-fold cross-validation.

We depict the distribution of the fitted LGDs from NN in Fig. 6. Although some fitted LGDs are outside the range of [0, 1], these numbers constitute a fairly small proportion (less than 8%) of the sample. The distribution of the fitted LGDs is slightly bi-modal, with higher concentrations in the intervals of [0, 0.1) and [0.7, 0.8). This result further confirms our earlier finding that the distribution of fitted LGDs is not directly related to model performance. So the bi-modal LGD distribution should be of only secondary concern in LGD modeling.

The major drawback of the NN method is that the model parameters are not uniquely identified and there is no straight forward way to show or test the relationship between the dependent and explanatory variables. For illustrative purpose, we show in Fig. 7 plots of fitted LGDs from the in-sample estimated NN model with $n = 25$ against 20 observed seniority index and industry distance-

[14] Note that Table 7 shows occasional deviations away from the increasing or decreasing trend in R-squared as n increases. This is likely due to the multi-modal error surface of the NN – even with the same n the optimization algorithm may converge to different local minima with different random starting values of the model parameters. To avoid local minima, we estimate the NN model ten times using ten different sets of random starting values and the model with the best fit is used.

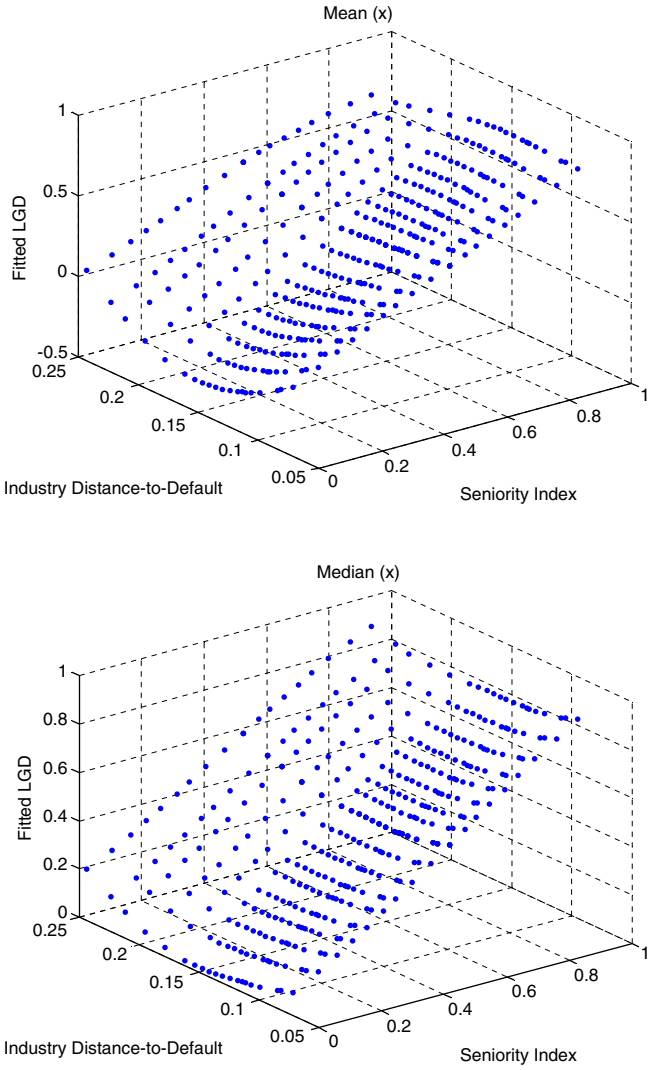


Fig. 7. Fitted LGDs from the neural network plotted against seniority index and industry distance-to-default.

to-default ranging from the lowest to the highest values, holding the values of the rest of the explanatory variables at the sample mean (the top panel) or the sample median (the bottom panel).

We can observe the following from Fig. 7. First, there is a positive relationship between LGD and seniority index, holding the rest of the explanatory variables at either the sample mean or the sample median. This is consistent with the positive correlation reported in Panel B of Table 2, and the results in Tables 3 and 5 and Fig. 4. Second, the impact of seniority index on LGD is stronger when industry distance-to-default is smaller (say under 15%) than when industry distance-to-default is larger (say above 20%). This finding suggests that relative standing among creditors is more important when industry condition is more severe. Third, contrary to the results from the parametric models, the relationship between LGD and industry distance-to-default captured by NN varies with the level of seniority index and is non-monotonic. LGD does not seem to vary much with industry distance-to-default when seniority index is high (say close to 1) and when industry distance-to-default is under 0.15. But LGD becomes positively correlated with industry distance-to-default when seniority index is low (say under 0.5) and when industry distance-to-default is above 0.15. Therefore, industry condition mainly affects junior creditors, indicating that recovery by senior creditors do not rely heavily on collateral sales.

5. Further investigations

5.1. Model performance on revolvers and term loans

We also investigate the performance of these six modeling methods on a smaller sample (1355 observations) consisting of revolvers and term loans. Results from this sample should be more relevant to banks' portfolio and the smaller sample size can also test the robustness of our earlier findings. We find that our conclusion does not change. The goodness-of-fit from the non-parametric methods exceeds that from the parametric methods. Second, the FRR method tends to produce slightly better goodness-of-fit than the OLS method. Third, performance of the transformation methods is quite sensitive to the choice of ε, and the optimal ε is still around 0.05 for this sample. We have also tried the same methods on the remaining instruments consisting of bonds. We find that our conclusions hold in this alternative sample as well.

5.2. Model performance when firm-level variables are included in the model

In the results reported so far, we do not include firm-level variables among the RHS variables. Incorporating these variables significantly reduces the sample size to 1267 observations. When we try the same methods on this smaller sample, our conclusions about the comparative performance of the various methods to model LGDs do not change. However, we caution that this conclusion might be specific to the nature of our sample (i.e., predominantly bankruptcy cases and large firms) and this question is worth further examination on alternative samples in the future.

5.3. Model performance when only dummy variables are included in the model

Further, in separate analysis, we find that non-parametric methods do not enjoy any advantage relative to parametric methods when the explanatory variables consist solely of the instrument type and collateral-type variables – the explanatory variables commonly used by banks in modeling LGD.[15] There are two reasons for the inferior performance of the non-parametric methods in this case. First, in models with only instrument type and collateral-type variables, the independent variables are all 0/1 dummy variables, with no non-linearity in the model. Consequently, non-parametric methods do not have any edge. Second, unlike the non-parametric methods, an intercept term is included in the parametric methods, which results in slightly better fit. We therefore conclude that the advantage enjoyed by non-parametric methods relative to parametric methods stems from their ability to model the non-linear relation between LGDs and continuous variables. Therefore, unless continuous variables are included in the model, non-parametric methods do not outperform parametric methods, even with a very large sample size. Modelers of LGD should thus try to compare performance of alternative parametric and non-parametric methods on their own data and the set of available explanatory variables to determine which method to choose.

[15] If we require a minimum of 5 observations at each leaf, the regression tree produces an R-squared of 0.322 and SSE of 373.77, and the NN produces R-squared of 0.322 and SSE of 373.83. These are worse than OLS (with an R-squared at 0.370 and SSE at 347.07) and FRR (with an R-squared at 0.365 and SSE at 349.98). With $\varepsilon = 0.05$, we get R-squared = 0.353 and SSE = 356.54 for IGR, and R-squared = 0.352 and SSE = 357.17 for IGR-BT.

6. Conclusions

This paper compares various parametric and non-parametric methods to model and forecast LGD, a continuous random variable that lies in the interval of [0, 1] and often follows a bi-modal distribution. We include in this study four parametric methods, namely, the OLS, fractional response, inverse Gaussian, and inverse Gaussian with beta transformation regressions, and two non-parametric methods, namely, the neural network and regression tree.

We find that the non-parametric methods outperform the parametric methods in terms of model fit and predictive accuracy. The neural network method delivers very good model fit both in sample and in the 10-fold cross-validation. One of the limitations of the neural network method is that the estimated model is often considered as a "black box," since there is no straight forward way to show the complex non-linear underlying relationships. The regression tree method also provides very high predictive accuracy. After a decent number of splits, the goodness-of-fit from this method exceeds those from the parametric methods. When we reduce the minimum size requirement to 70 or less and allow 41 or more splits, the regression tree method provides better fit than the neural network in the 10-fold cross-validation. Further, the regression tree does not appear to have the over-fitting problem even when the minimum size requirement is reduced to 5 with as many as 342 splits, although such a large tree may not be practical in a production environment.

Among the parametric methods, fractional response regression tends to produce slightly better goodness-of-fit than the OLS method, and both the OLS and the fractional response regression methods are able to provide decent model fit. Both transformation methods are very sensitive to the choice of ε (which is needed to transform LGD = 0 or 1). With an optimal ε, their performance can match that of the OLS regression. Therefore, users of these transformation methods need to exercise great caution.

Further, we find that the bi-modal distribution may be of only secondary concern when modeling LGD. Finally, inferences on some instrument-level variables are slightly different between the models, suggesting that the instrument-level variables may not have a clear-cut relation with LGD.

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